

cse 438 - robotics

SMART PLANT MONITORING BOT

Presented By :
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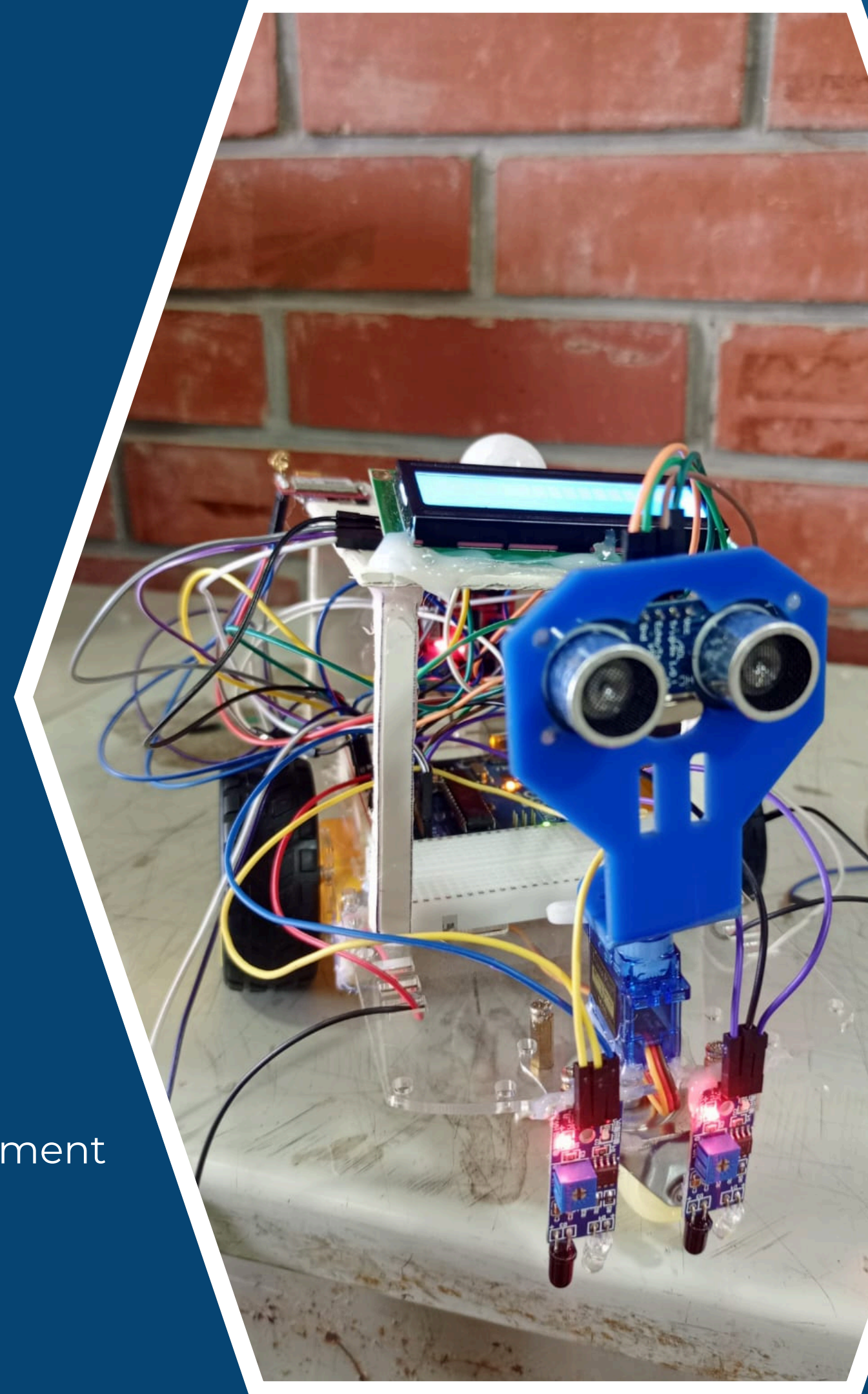
Presented To :
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Professor Department of
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Executive Summary

Smart Plant Monitoring Bot is an Arduino-based robot designed to monitor and protect plants automatically. It follows a black line around the plant area, detects obstacles, and gives alerts when any unwanted movement is detected near the plants.

The bot is capable of:

- Following a black line using dual IR sensors
- Autonomously patrolling the plant area without manual control
- Detecting and avoiding obstacles using an ultrasonic sensor on a servo
- Resuming line patrol automatically after obstacle avoidance
- Detecting nearby motion using a PIR sensor
- Triggering a continuous buzzer alarm during threat presence
- Displaying live status and threat distance on an I2C LCD screen
- Sending an SMS alert with intruder distance to the owner via GSM
- Protecting plants from cats, dogs, birds, insects and unauthorized movement



Contribution

Hardware Contribution

- Chassis and 2-wheel drive assembly
- Component selection and procurement
- L298N motor driver wiring and calibration
- IR sensor connection and line sensitivity tuning
- Servo motor installation and angle calibration (30°–150°)
- Ultrasonic sensor mounting on servo
- PIR sensor placement and range testing
- GSM SIM800L wiring and SMS testing
- I2C LCD 16×2 integration
- Power bank connection via Arduino USB
- Full system integration and field testing

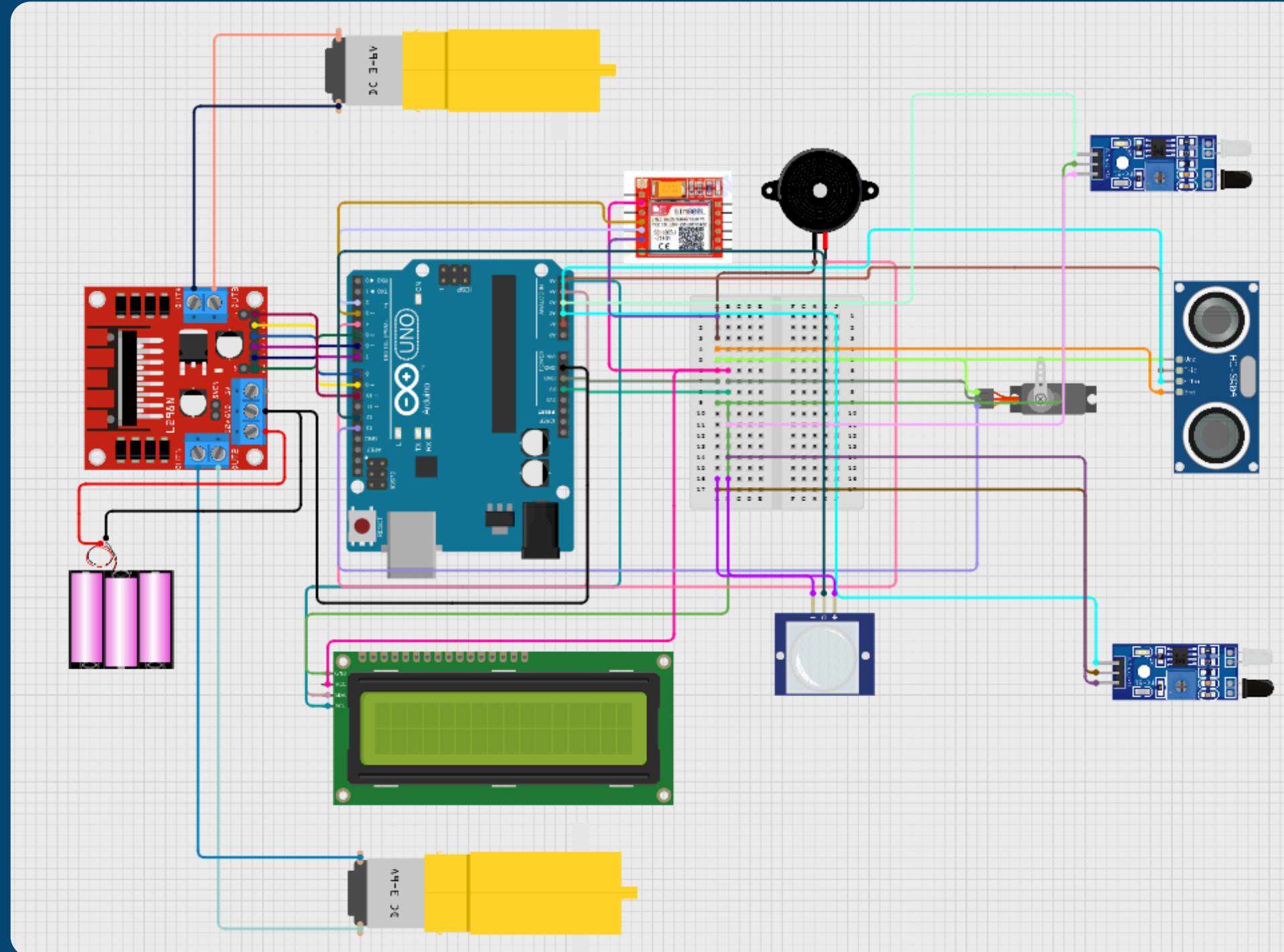
Survey Contribution

- Research on plant damage by cats, dogs, birds and insects
- Study on unattended home plant damage without monitoring
- Analysis of automated bots for 24/7 plant protection
- Survey on common intruder threats to home plant areas
- Research on real-time SMS alerts for absent plant owners
- Study on low-cost Arduino solutions for plant protection
- Literature review on line-following robots for garden use
- Analysis of early threat detection preventing plant damage

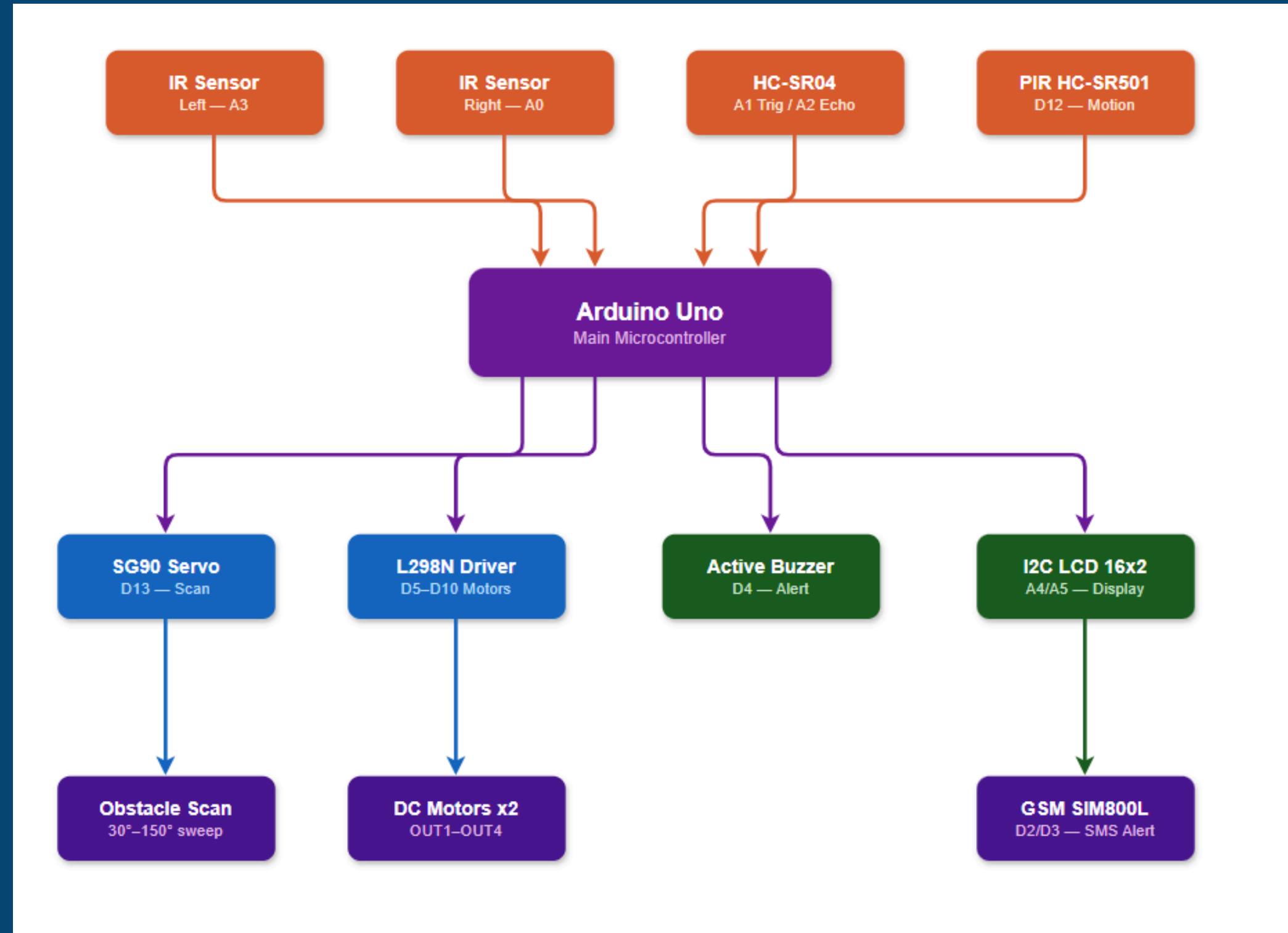
Equipment & Components

Component	Quantity	Purpose
Arduino Uno	1	Main microcontroller for robot control
IR Sensor	2	Detects black line for line
Ultrasonic Sensor	1	Measures distance and detects obstacles
Servo Motor	1	Rotates the ultrasonic sensor for path scanning
PIR Sensor	1	Detects motion near the plants
GSM Module	1	Sends SMS alerts to the owner
16x2 I2C LCD	1	Displays status messages and
L298N Motor Driver	1	Controls the DC motors
DC Gear Motor	2	Drives the robot wheels
Buzzer	1	Provides sound alerts
Mini Chassis with Wheels	1	Supports robot movement and structure
Battery / Power Supply	1	Supplies power to the system

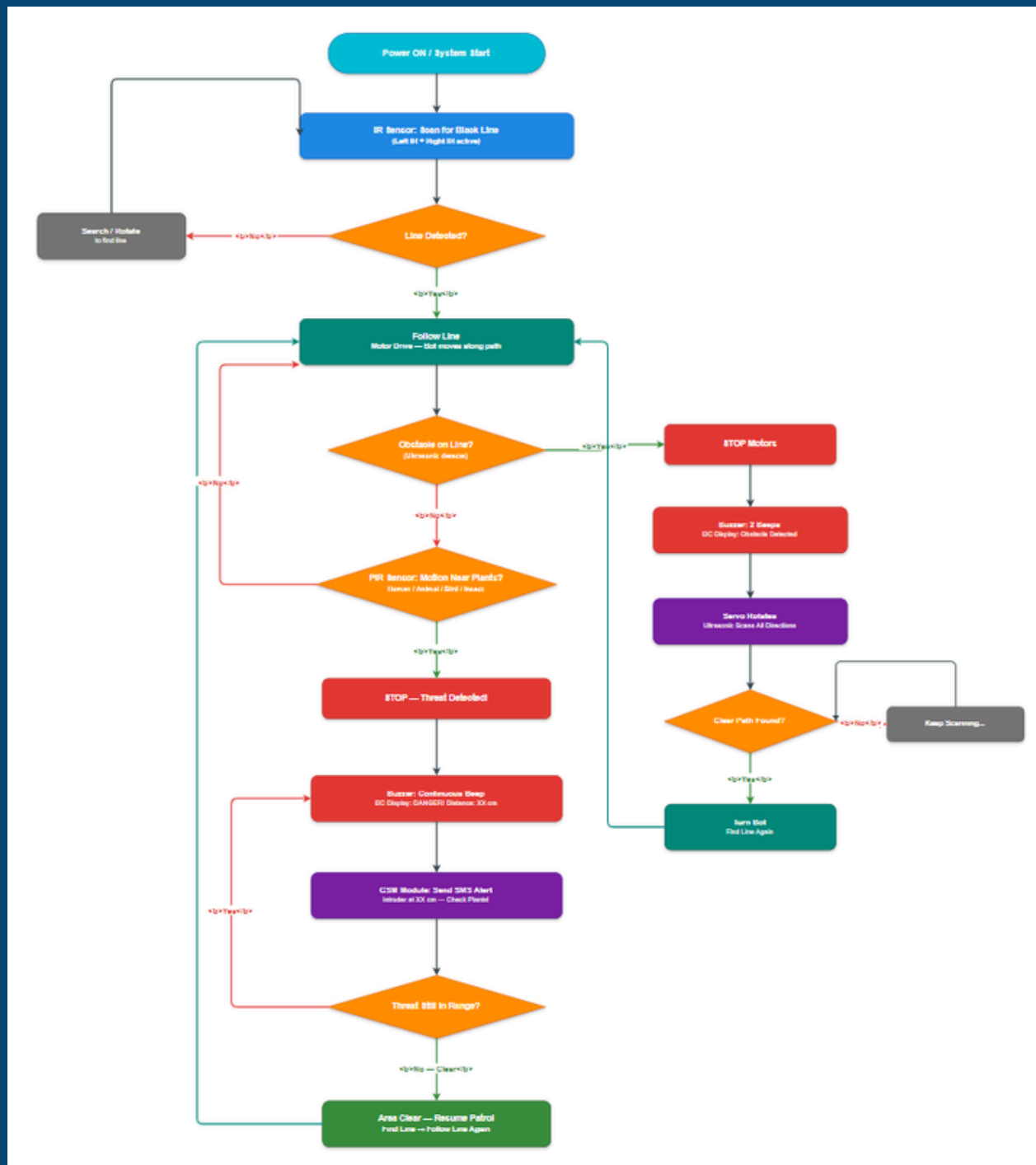
Circuit Diagram



Embedded System Architecture



Flow Diagram



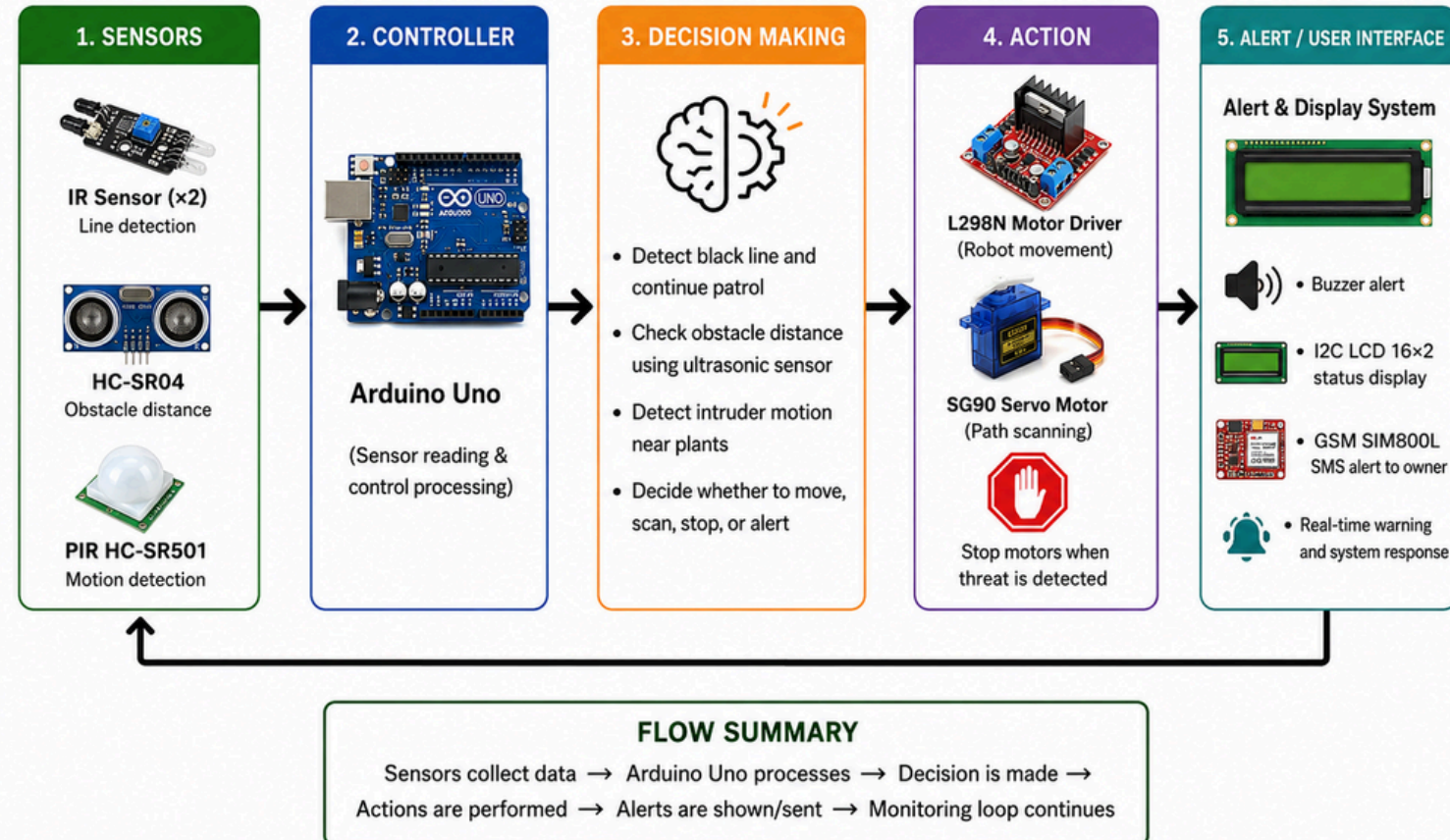
Connections

- Arduino Uno → L298N Motor Driver → DC Motors
- Arduino Uno → IR Sensors → A3, A0
- Arduino Uno → Ultrasonic Sensor → A1, A2
- Arduino Uno → Servo Motor → D13
- Arduino Uno → PIR Sensor → D12
- Arduino Uno → Buzzer → D4
- Arduino Uno → I2C LCD 16×2 → A4/SDA, A5/SCL
- Arduino Uno → GSM SIM800L → D2, D3

Interaction Flow

INTERACTION FLOW DIAGRAM

Smart Plant Monitoring Bot
Plant Protection & Monitoring System



- Real-Time Plant Protection Monitoring
- IR-Based Line Following Patrol
- Ultrasonic Obstacle Detection & Servo Scanning
- PIR-Based Danger Alert with LCD, Buzzer & GSM SMS

Related Work

Line-following robots using IR sensors can autonomously navigate predefined paths for plant monitoring. (Line Following Robot for Agriculture, 2022)

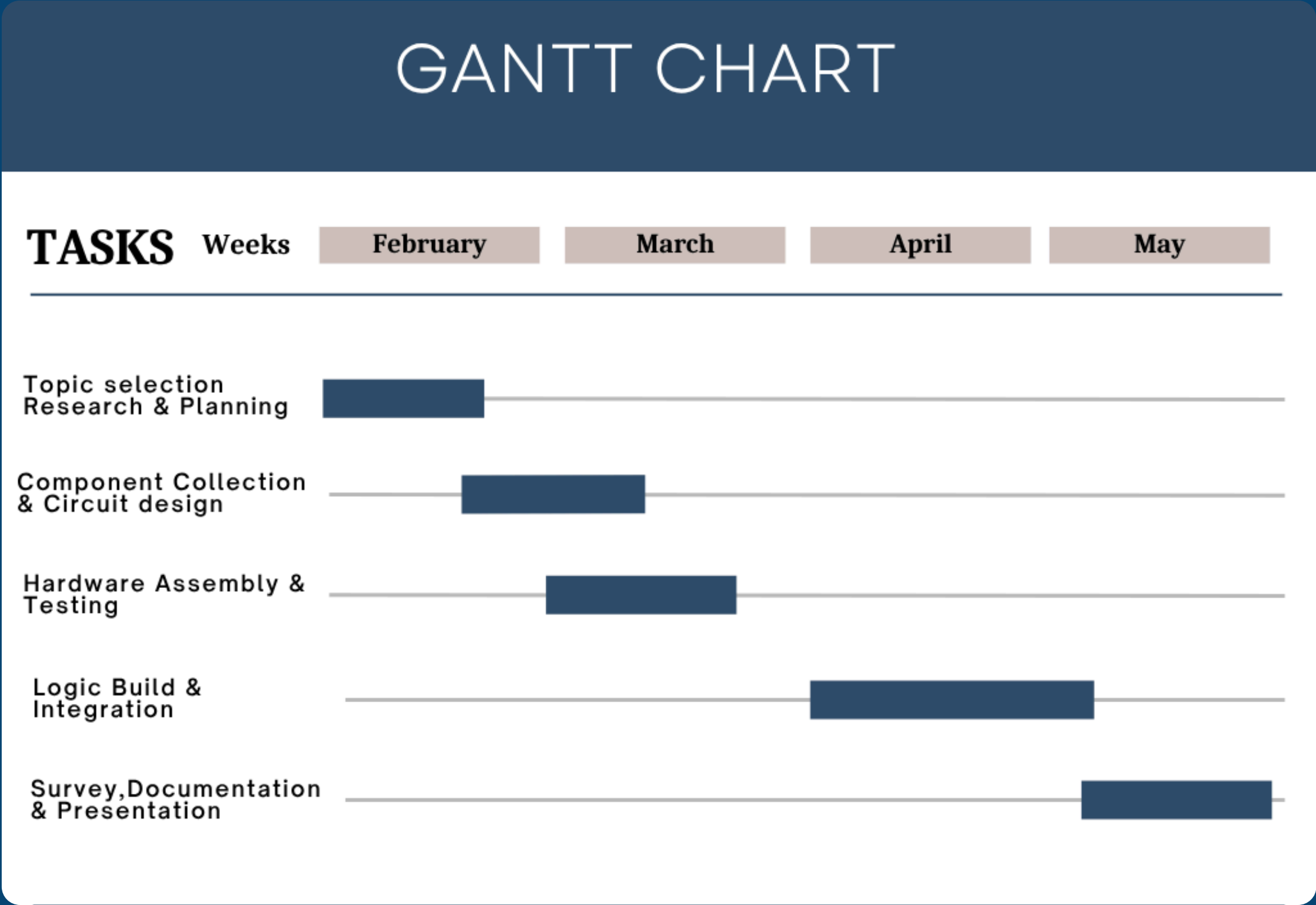
Arduino Uno effectively supports autonomous patrol and alert systems using PIR sensors and GSM modules for real-time intrusion detection. (IoT Based Plant Protection System, 2023)

Ultrasonic sensors mounted on servo motors significantly improve obstacle detection and path recovery accuracy in autonomous rovers. (Obstacle Avoiding Agricultural Rover, 2021)

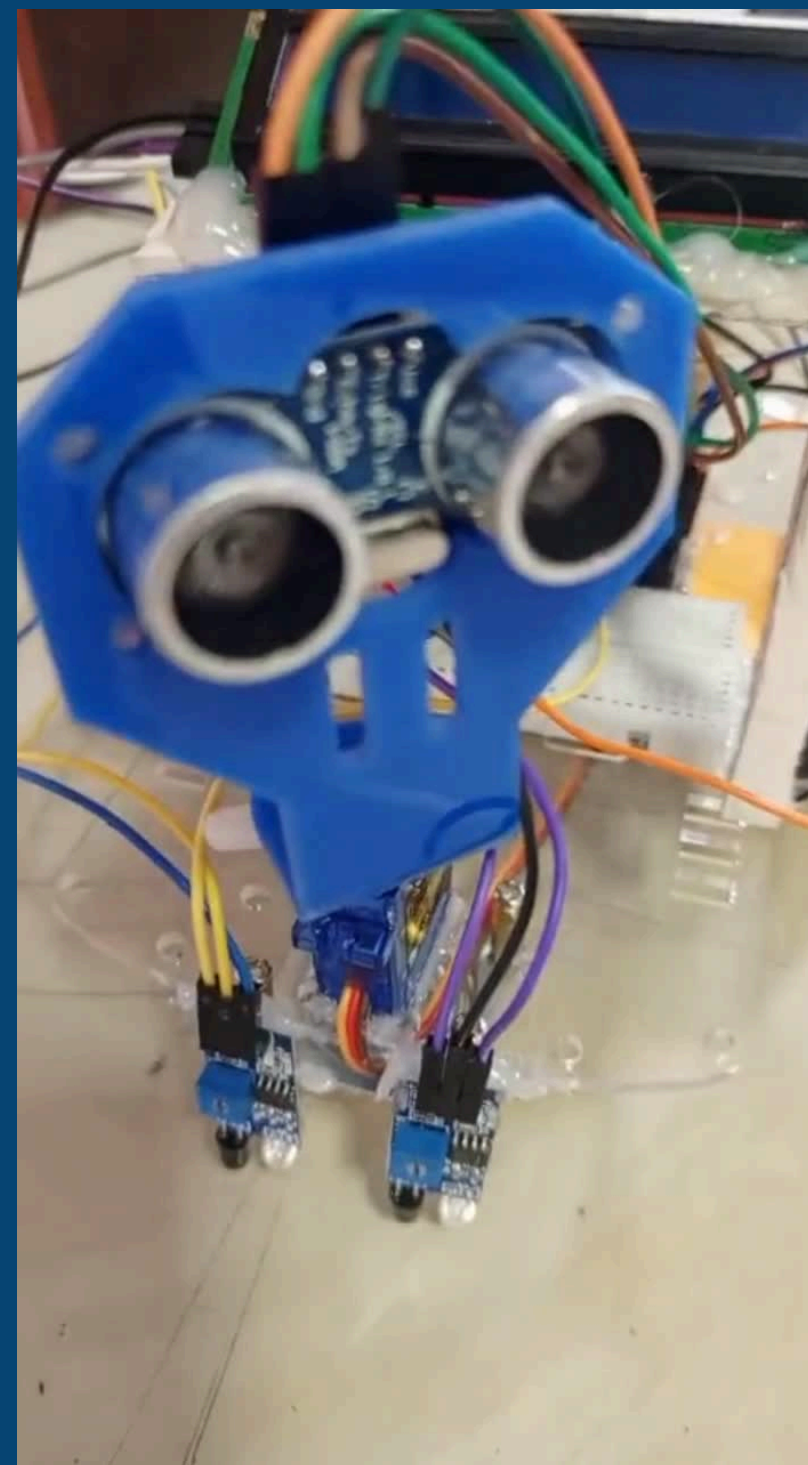
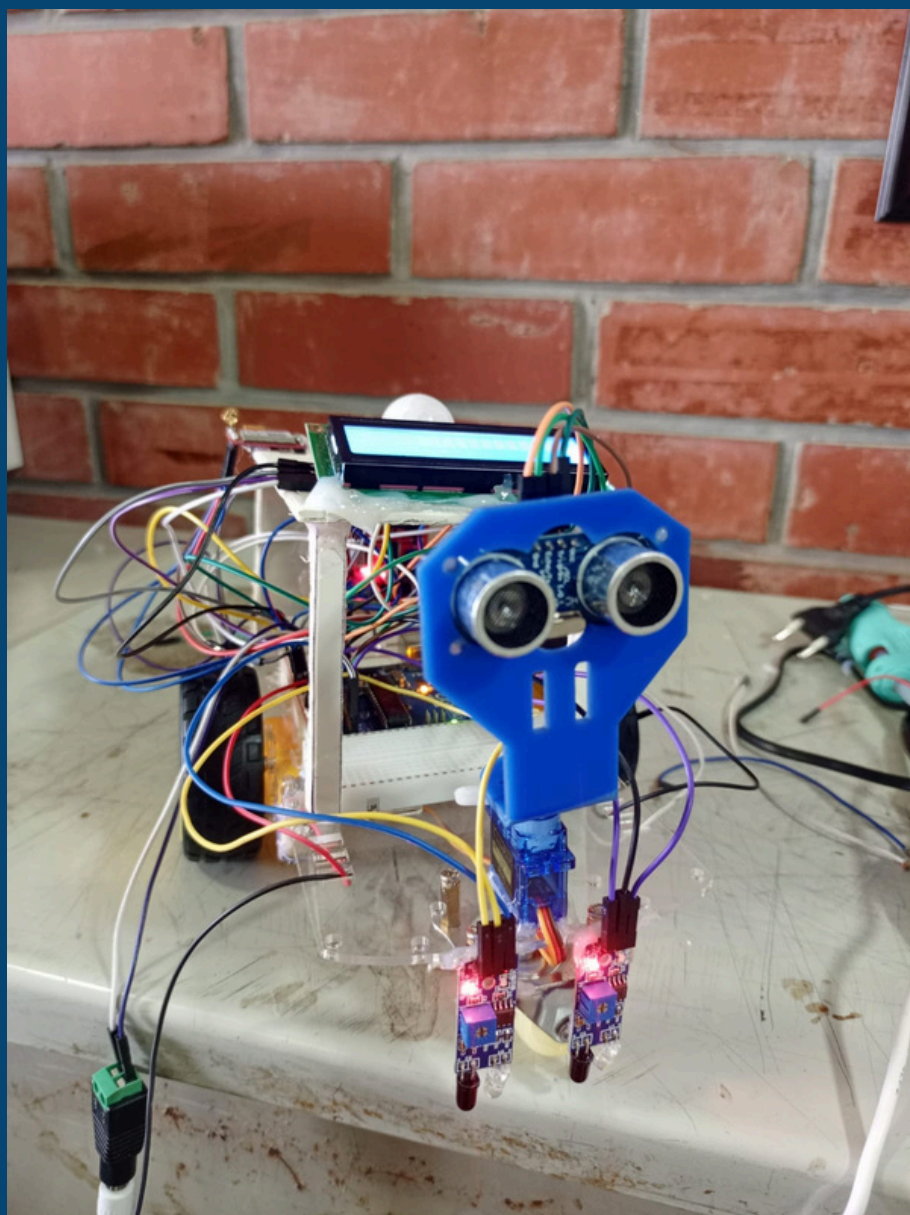
Our Contribution

- Autonomous line-following patrol bot for plant area monitoring
- Real-time intruder detection using PIR motion sensor with GSM SMS alert including live distance data
- Obstacle detection and automatic path recovery using servo-mounted ultrasonic sensor
- I2C LCD real-time status display for system feedback
- Low-cost Arduino-based solution suitable for small gardens and farms in Bangladesh
- Continuous autonomous patrol without human presence required

Gantt Chart



Demo Video





THANK YOU



ANY QUESTION ?